Weligama Coconut Leaf Wilt Disease (WCLWD) Severity Assessment using ABCD Rule

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Project Proposal Report

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DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

One of the dangerous diseases impacting Sri Lankan coconut fields is Weligama Coconut Leaf Wilt Disease (WCLWD). Visible signs of the disease include leaf flaccidity, uneven yellowing, and browning of the leaf tips, which progressively lower coconut yield and productivity. For the coconut industry to manage diseases and avoid significant financial loses, early detection and severity assessment are essential. Existing approaches to disease identification mostly rely on laboratory-based procedures including molecular diagnostics, PCR, and ELISA. These techniques are costly, time consuming, and unsuitable for extensive field surveillance even if they offer precise pathogen detection. Though many of these systems rely on image level classification and do not provide accurate severity assessment, recent image based deep learning techniques have improved disease identification. Additionally, segmentation requires human pixel level annotation, which is time consuming and needs specialized knowledge. Python based deep learning frameworks like PyTorch and image processing libraries like OpenCV will be used to create the system. A scalable and automated framework that can precisely determine the severity of WCLWD from field photos is the expected result. By promoting early disease detection and helping researchers and plantation managers make decisions, this study supports sustainable agriculture.

Keywords: *WCLWD, Weakly Supervised Segmentation, ABCD Rule, Deep Learning, Coconut Disease Detection*

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LIST OF ABBREVIATIONS

Table 1: List of Abbreviations

Abbreviation	Full Form
WCLWD	Weligama Coconut Leaf Wilt Disease
ABCD	Asymmetry, Boundary, Colour, Distribution
WSSS	Weakly Supervised Semantic Segmentation
ViT	Vision Transformer
Grad-CAM	Gradient-weighted Class Activation Mapping
DSI	Disease Severity Index
CNN	Convolutional Neural Network
PCR	Polymerase Chain Reaction
ELISA	Enzyme-Linked Immunosorbent Assay
IoU	Intersection over Union
F1-score	Harmonic Mean of Precision and Recall
AI	Artificial Intelligence
ML	Machine Learning
CRI	Coconut Research Institute
GPU	Graphics Processing Unit

1. INTRODUCTION

1.1 Background and Context

One of Sri Lanka's main agricultural industries and a big contributor to the country's economy is coconut cultivation. However, diseases that decrease yield and plantation productivity are becoming more common on coconut plantations. WCLWD is currently one of the most serious diseases impacting coconut trees in southern Sri Lanka.

WCLWD causes progressive damage to coconut leaves, characterized by symptoms such as flaccidity, uneven yellowing, and browning of leaflet tips [1]. These symptoms cause the coconut palm to become weaker and less able to produce nuts, which eventually results in large yield losses. According to studies, if WCLWD is not identified and treated early on, it can seriously harm coconut plantations [1].

Conventional WCLWD diagnostic approaches depend on laboratory technologies like ELISA based immunological detection, nested PCR [2] [3]. These techniques are accurate in identifying the presence of phytoplasma linked to the disease, but they require expensive equipment, laboratory space, and skilled personnel [4].

Recent studies explored the application of deep learning and artificial intelligence to the identification of plant diseases through image processing techniques [5]. These methods facilitate quicker diagnosis by automatically identifying disease symptoms from leaf photos. Nevertheless, a lot of current methods merely determine whether a plant is healthy or diseased; they are unable to assess the severity of the disease.

Accurate disease severity assessment is important because it helps determine the level of infection and guides appropriate management strategies. Therefore, farmers, researchers, and agricultural authorities would greatly benefit from an automated approach that uses field photos to identify and assess the severity of WCLWD.

1.2 Literature Review

Several studies have looked into various approaches to identifying WCLWD and related plant diseases. The phytoplasma linked to WCLWD infection has been identified using molecular detection techniques including nested PCR [2]. Similarly, phytoplasma DNA has been found in infected coconut palms with great diagnostic accuracy using real-time PCR techniques [3]. Nevertheless, these methods cannot be used for real-time field level monitoring and need laboratory facilities.

WCLWD associated phytoplasma have also been identified using immunological detection techniques as indirect ELISA [4]. Although these methods offer accurate pathogen detection, they are costly, time consuming, and require specific knowledge.

Studies explored the effects of WCLWD on coconut palm shape, physiology, and yield in addition to laboratory-based detection techniques, highlighting the significance of early diagnosis and monitoring [1]. The requirement for efficient disease monitoring systems has been further highlighted by research on insect vectors that spread the disease [6]. Additionally,

assessments of hybrid coconuts have been carried out to find varieties that show resistance to WCLWD infection [7].

More recently, deep learning techniques have been applied to automated image processing approaches for plant disease detection. For instance, WCLWD and coconut caterpillar infestations have been identified from leaf photos using models based on convolutional neural networks [5]. While these models show a high accuracy in classifying diseases, they are more concerned with detecting the existence of disease than evaluating its severity.

Another limitation of many deep learning approaches is the requirement for manual pixel-level annotation for segmentation-based disease analysis. Manual annotation is costly, time-consuming, and requires expert knowledge [8]. When working with huge image databases, this constraint makes scaling such systems challenging.

Table 2: Comparison of Existing Approaches vs Proposed Method

Approach	Technique Used	Main Purpose	Strengths	Limitations	Proposed Method Improvement
Molecular Detection Methods [2] [3]	Nested PCR, Real-time PCR	Detect phytoplasma associated with WCLWD	Highly accurate pathogen detection	Require laboratory facilities, expensive equipment, not suitable for field monitoring	Performs image-based disease severity assessment directly from leaf images in field conditions
Immunological Detection [4]	Indirect ELISA	Detect WCLWD associated phytoplasma	Reliable laboratory diagnosis	Time consuming, costly, requires specialized expertise	Provides faster automated detection without laboratory tests
Disease Impact Studies [1]	Morphological and physiological analysis	Study the impact of WCLWD on coconut palms	Helps understand disease symptoms and yield loss	No automated detection or monitoring system	Automatically detects and quantifies disease severity
Vector Identification Studies [6]	Entomological research	Identify insect vectors spreading WCLWD	Helps understand disease transmission	Does not support detection or severity assessment	Focuses on automated disease severity monitoring
Hybrid Tolerance Studies [7]	Hybrid evaluation	Identify coconut varieties	Supports disease	Does not provide real-time	Enables early severity estimation

		tolerant to WCLWD	management strategies	disease detection	for disease management decisions
Deep Learning Disease Detection [5]	CNN-based image Classification	Detection of the presence of WCLWD and pest infestation	Automated detection using leaf images	Focuses on classification only, not severity estimation	Performs both disease localization and severity estimation
Pixel-level Annotation Methods [8]	Segmentation with manual pixel labeling	Estimate plant disease severity	Accurate segmentation of diseased regions	Manual annotation is expensive, time-consuming, and expert-dependent	Uses WSSS eliminating manual annotations
Proposed Method	Vision Transformer + Grad-CAM + WSSS + ABCD Rule	Automated WCLWD severity assessment	Automated detection, segmentation, and severity scoring	-	Provides scalable, annotation-efficient, and field-deployable disease severity assessment system

1.3 Research Gap

There are a number of methods for identifying WCLWD and other plant diseases, however as of right now, there isn't a scalable automated system that can reliably determine the severity of the disease from field images. The majority of current systems do not offer quantitative severity estimation and instead concentrate solely on disease identification [5].

Additionally, segmentation-based disease analysis usually requires human pixel-level annotation, which is costly, time-consuming, and heavily reliant on expert participation [8]. The practicality of implementing such systems in actual agricultural settings is reduced by these constraints.

Furthermore, it is difficult to use laboratory-based detection techniques like PCR and ELISA for real-time monitoring in coconut plantations [2] [4]. A field-deployable automated system that can identify and quantify disease severity directly from leaf images is therefore required.

In order to close this gap, this study suggests a weakly supervised semantic segmentation method in combination with the ABCD morphological severity assessment rule to enable automated severity estimation of WCLWD symptoms from images of coconut leaves.

1.4 Research Problem

One of the most serious diseases affecting coconut cultivation in Sri Lanka, especially in the southern parts of the country, is WCLWD. Through symptoms like leaf flaccidity, yellowing, and browning of leaflet tips, the disease progressively weakens coconut palms, ultimately resulting in decreased coconut production and productivity [5]. To prevent the disease from spreading and reduce financial losses on coconut plantations, early detection and continuous monitoring of the disease's severity are important.

Although a number of laboratory-based methods, including nested PCR and ELISA, have been developed to detect the phytoplasma linked to WCLWD, these methods are not practical for large-scale field monitoring because they require specialized equipment, laboratory facilities, and trained personnel [2] [4]. Therefore, in actual plantation settings, these approaches are unable to offer a practical and scalable solution for continuous disease surveillance.

Recent studies explored the application of computer vision and deep learning methods for leaf image-based plant disease identification. Automated disease detection in agricultural applications is made possible by the great accuracy with which CNN models have identified plant diseases from images [5] [9]. Nevertheless, the majority of these models are restricted to binary or multi-class categorization, whether a plant is healthy or infected without offering a thorough assessment of the severity of the disease.

Furthermore, segmentation-based methods that seek to locate diseased regions frequently requires manual pixel-level annotation, which is labor-intensive, time-consuming, and reliant on expert knowledge [8]. Applying such systems to large image datasets or real-world agricultural monitoring settings is challenging due to these constraints. Recent developments in WSSS methods have demonstrated potential for reducing annotation requirements while maintaining accurate disease regions localization in images [10].

Therefore, an automatic system that can identify diseased regions and determine the severity of the disease immediately from images of coconut leaves without requiring a lot of manual annotation is obviously needed. Resolving this issue will facilitate more effective disease surveillance, promote early detection methods to prevent the development of WCLWD in coconut plantations, and help agricultural researchers and plantation managers make decisions.

2. OBJECTIVE

2.1 Main Objective

Use WSSS alongside with the ABCD severity assessment rule to provide an automated framework for WCLWD identification and quantitative severity evaluation in real world conditions.

2.2 Specific Objectives

- Develop a weakly supervised semantic segmentation model that can identify WCLWD-affected areas in images of coconut leaves without the need for manual pixel-level annotation.
- Utilize weak supervision and image-based disease symptoms to precisely identify and segment diseased areas from input leaf images.
- Extract lesion morphological features such as asymmetry, border irregularity, colour variation, and diameter factors, from the segmented diseased regions using the ABCD rule.
- Design a quantitative technique for assessing the severity of the disease by calculating severity levels using segmented lesion regions and extracted ABCD features.
- Analyze the proposed framework's applicability for real-world coconut plantation monitoring, compare its performance with other fully supervised techniques documented in the literature, and validate it using conventional segmentation and classification metrics.

3. METHODOLOGY

Since the main objective of the study is to design, develop, and test an automated system for determining the severity of WCLWD, this research will uses the DSR methodology. The objective of the research is to develop a useful artifact, an intelligent image-based severity assessment system and verify its effectiveness in real-world conditions.

An existing collection of images of coconut leaves that were previously gathered from coconut plantations will be used in this study. In order to ensure differences in image resolution, lighting settings, and background surroundings, the dataset includes images taken using a variety of devices, including DSLR cameras and smartphone cameras (such as iPhone models). Images of important WCLWD symptoms, such as leaf flaccidity, uneven yellowing, and leaflet tip browning, are included in the dataset. By enabling the model to learn disease patterns under a variety of real-world conditions frequently observed in plantation environments, the use of photos taken with various devices enhances the model's robustness.

These dataset will go through a preprocessing step to standardize image inputs prior to model training. OpenCV and Scikit-image will be used for image preparation, which includes like noise reduction, contrast enhancement, scaling, and normalization. Before the images are put into the deep learning models, these preprocessing procedures serve to enhance image quality, reduce differences brought on by various capture equipment, and guarantee consistency throughout the dataset.

A ViT model will be applied for both feature learning and disease detection. ViT was chosen because it is good at collecting fine-grained visual patterns and global image relationships, both of which are crucial for determining how leaf symptoms propagate. The model will generate discriminative features from input images and categorize visual patterns associated with diseases. Grad-CAM will be used to visualize the areas that most significantly affect the model's predictions in order to enhance interpretability. The coarse localization of infected areas will be provided by these activation maps.

WSSS will be used to accurately identify diseased regions. WSSS does not require manually annotated pixel-level masks, in contrast to conventional segmentation techniques. Rather, it produces pseudo-segmentation masks by combining Grad-CAM activation maps with image-level labels. This reduces the cost of annotation and increases the system's scalability. To enhance segmentation quality, further refinement methods, including thresholding, morphological procedures, and region filtering may be used.

Following the identification of diseased regions, the ABCD rule which stands for **Asymmetry, Boundary, Colour, and Distribution** will be used to extract morphological data.

- Asymmetry is a measure of how symmetrically or irregularly diseased regions are spread throughout the leaf.
- The form and irregularity of lesion edges will be assessed using boundary analysis.
- Using colour-based characteristics taken from the image, colour analysis will identify variations like yellowing and browning.

- The degree to which the symptoms are dispersed throughout the leaf surface will be measured by distribution.

OpenCV and Scikit-image image processing techniques, such as contour detection, shape analysis, colour histogram analysis, texture features, and region property measurements, will be used to extract these features. The disease symptoms will be quantitatively described by these extracted attributes.

To calculate the degree of infection, a DSI will be computed using the retrieved ABCD features. To determine the overall degree of infection, the DSI will integrate the segmented diseased regions and assessed lesion characteristics. The end product will be presented as a severity category, such as mild, moderate, severe or critical as well as a severity percentage. This severity rating offers a useful interpretation that can help plantation managers and agricultural researchers keep an eye on diseases.

Python will be the main programming language used to implement the system. The Vsion Transformer model will be created and trained using PyTorch, along with Grad-CAM and weakly supervised segmentation methods. For image preprocessing, segmentation refinement, and feature extraction, OpenCV and Scikit-image will be used. Numerical calculations and dataset management will be supported by additional libraries like NumPy and Pandas.

Finally, quantitative metrics will be used to assess the suggested framework's performance. IoU will be used to measure the segmentation performance and determine how well diseased regions are identified. The F1-score, which offers a balanced measure of precision and recall, will be used to assess the disease detection ability. By contrasting the expected severity scores with expert-assigned severity levels, the efficacy of severity estimation component will be verified. These assessment metrics guarantee that the suggested system is dependable and appropriate for facilitating actual WCLWD severity monitoring in coconut plantations.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

The proposed system consists of several interconnected components.

1. Image Input Module: Collects images of coconut leaves taken in the field.
2. Preprocessing Module: Performs image resizing, normalization, and noise removal.
3. Disease Detection Module: This module uses Vision Transformer models to identify features of diseases.
4. Segmentation Module: This module locates diseased regions using weakly supervised semantic segmentation.
5. Feature Extraction Module: Based on the ABCD rule, extracts morphological features.
6. Severity Calculation Module: Calculates the percentage and category od disease severity.

The quantitative severity score and severity classification will be the final output.

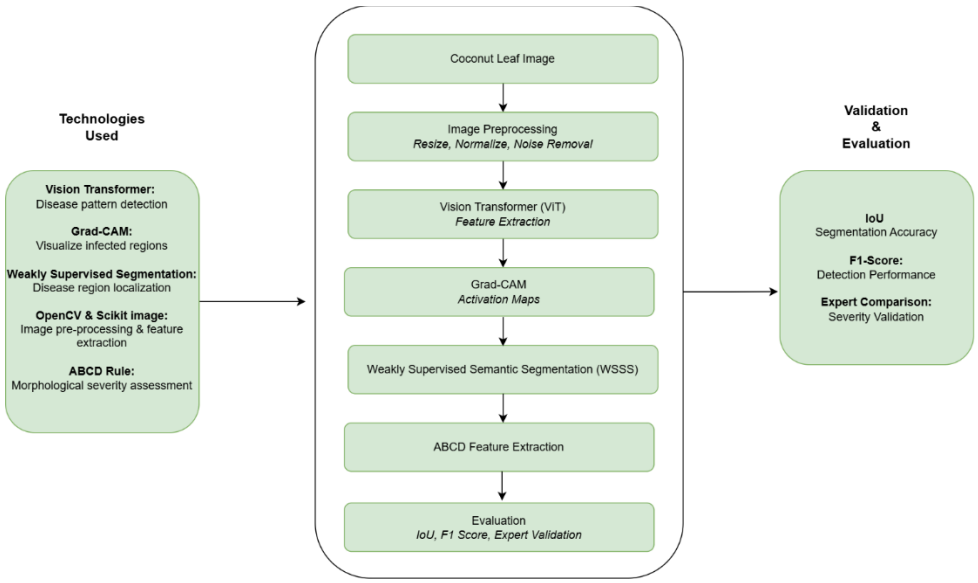


Figure 1:High Level System Architecture of Proposed System

5. USER REQUIREMENTS

Stakeholders:

- Coconut farmers
- Agricultural researchers
- Coconut Research Institute (CRI)
- Plantation managers

Functional Requirements

- Images of coconut leaves should be accepted as input by the system.
- Disease symptoms need to be automatically identified by the system.
- The system should identify diseased regions on the leaf.
- The percentage of disease severity should be determined by the system.
- The system should display disease severity categories.

Non-Functional Requirements

- High accuracy in disease detection
- Fast processing time
- Scalability for large datasets
- User-friendly interface

6. COMMERCIALIZATION PLAN

The suggested system will be developed as a web application to assist Sri Lanka's Coconut Research Institute (CRI) in tracking and evaluating the severity of Weligama Coconut Leaf Wilt Disease (WCLWD). The system's main goal is to offer an automated, image-based disease severity assessment tool that may help plantation managers, agricultural officers, and researchers detects infected coconut palms and analyze the disease's severity.

CRI researchers, agricultural extension agents, managers of coconut plantations, and government agricultural authorities involved in plantation management and coconut disease monitoring are among the system's target users. The technology can enhance plantation management and disease control decision-making processes by offering an automated platform for disease severity analysis.

The system's ability to automatically evaluate images of coconut leaves and generate disease severity levels is what makes it valuable because it eliminates the need for expensive manual inspections and lab-based diagnosis processes. This can enable quicker reaction to disease outbreaks and greatly increase the effectiveness of disease surveillance programs.

Initially, CRI may use the web application internally for monitoring and study. The system may eventually be expanded to provide extensive agricultural surveillance throughout Sri Lanka's coconut farms. Offering the platform as a subscription-based service to government agencies, plantation businesses, and agricultural organizations are possible commercialization prospects.

In order to support smart agriculture projects, the system can also be connected with other agricultural data platforms, allowing for long-term disease surveillance and data-driven plantation management. The suggested approach could help modern coconut disease surveillance in Sri Lanka by offering a scalable and easily accessible digital solution.

7. BUDGET AND JUSTIFICATION

A number of resources are needed for the proposed study, including storage for image collections, transportation for field visits, computing power for deep learning model training, and basic research costs. The estimated budget required for conducting the research is summarized in Table 3.

Table 3: Budget Estimation

No.	Expense Category	Description	Estimated Cost (LKR)
1	Transportation	Travel expenses from SLIIT Malabe to CRI Lunuwila for meetings, consultations, and dataset verification (multiple visits)	12,000
2	Data Collection & Field Support	Minor expenses related to field visits such as local transportation within plantations and coordination with CRI staff	3,000
3	External Storage Device	Portable 1TB external hard drive or SSD for storing image datasets, model checkpoints, and experimental results	10,000
4	Computing Resources	GPU usage or cloud computing credits required for training ViT and segmentation models	15,000
5	Miscellaneous	Unexpected research-related expenses such as software utilities or additional storage	3,000
Total Estimated Budget			43,000

Justification:

For research discussions, dataset verification, and consultation with agricultural specialists, transportation costs must be covered from SLIIT Malabe to CRI in Lunuwila. Throughout the research, several visits can be required.

Small logistical expenses incurred during plantation visits and research coordination activities are covered by field support expenses.

The coconut leaf image dataset, preprocessed images, training model weights, and experimental outputs produced throughout the deep learning model development process must all be stored on an external storage device.

WSSS models and deep learning models like ViT require a lot of processing power. These resources could be cloud-based training platforms or GPU-enabled environments.

Finally, a small miscellaneous allocation is included to cover unexpected expenses that may arise during the research period.

8. WORK BREAKDOWN STRUCTURE (WBS)

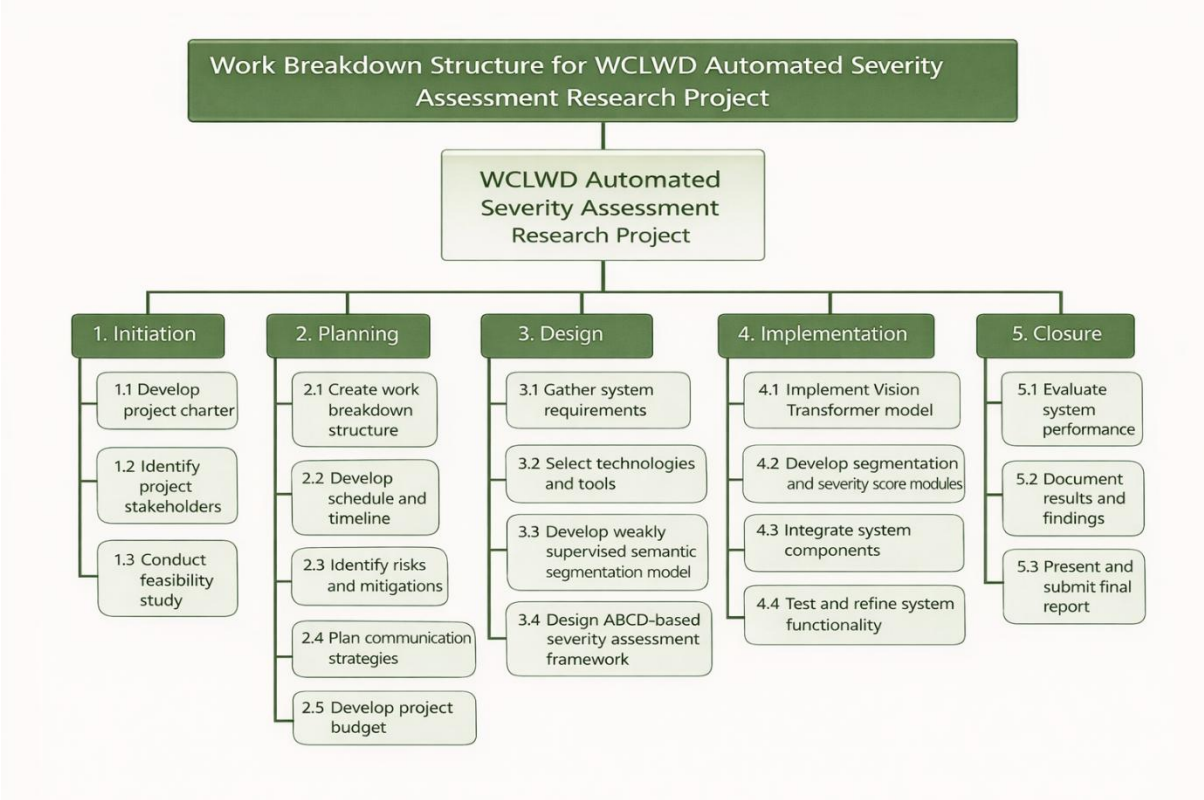


Figure 2: Work Breakdown Structure of Proposed System

9. GANTT CHART

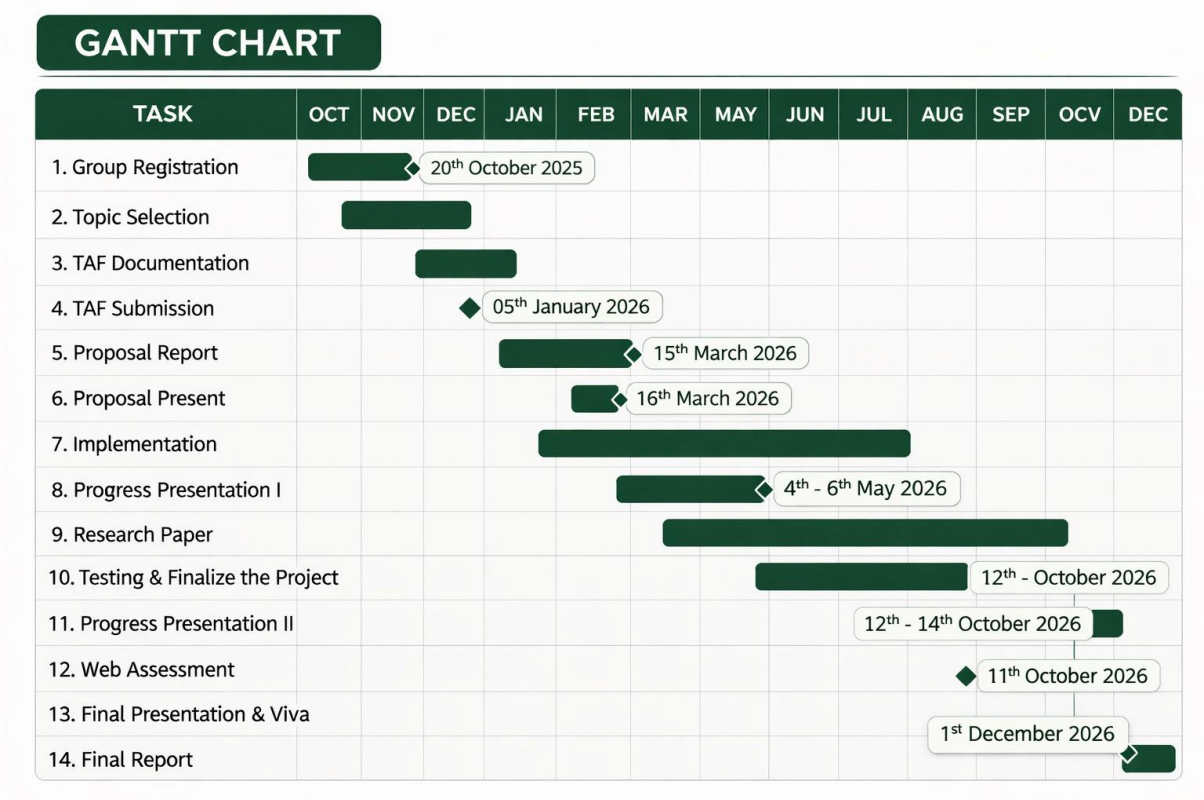


Figure 3: Gantt Chart

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